

DS Written Homework 2

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1 Binary Tree

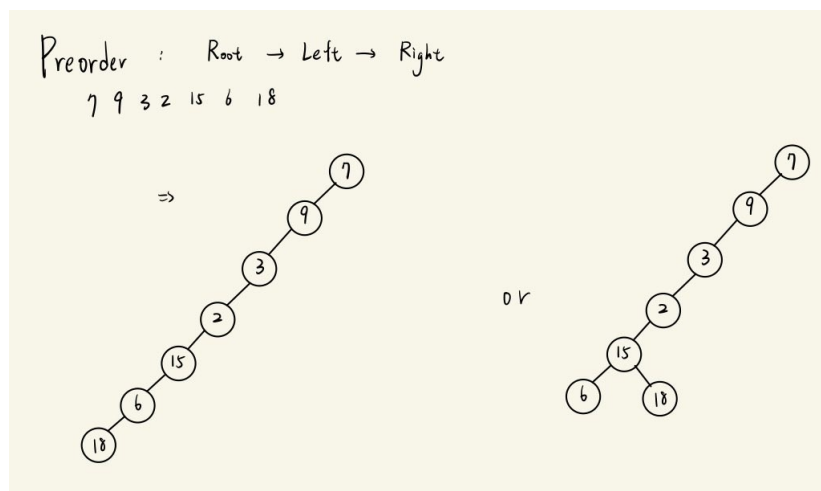


Figure 1: Binarytree

(a)

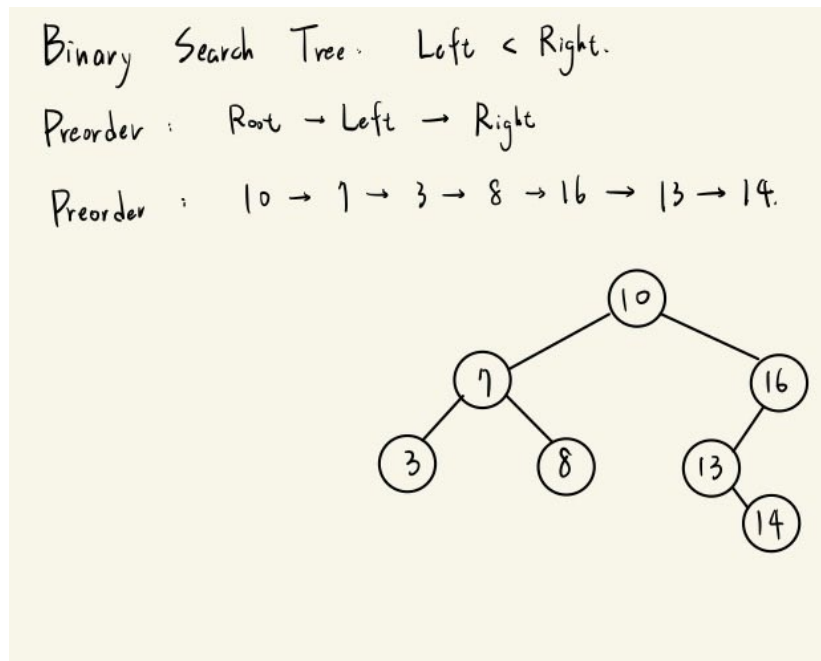


Figure 2: Binarytree

(b)

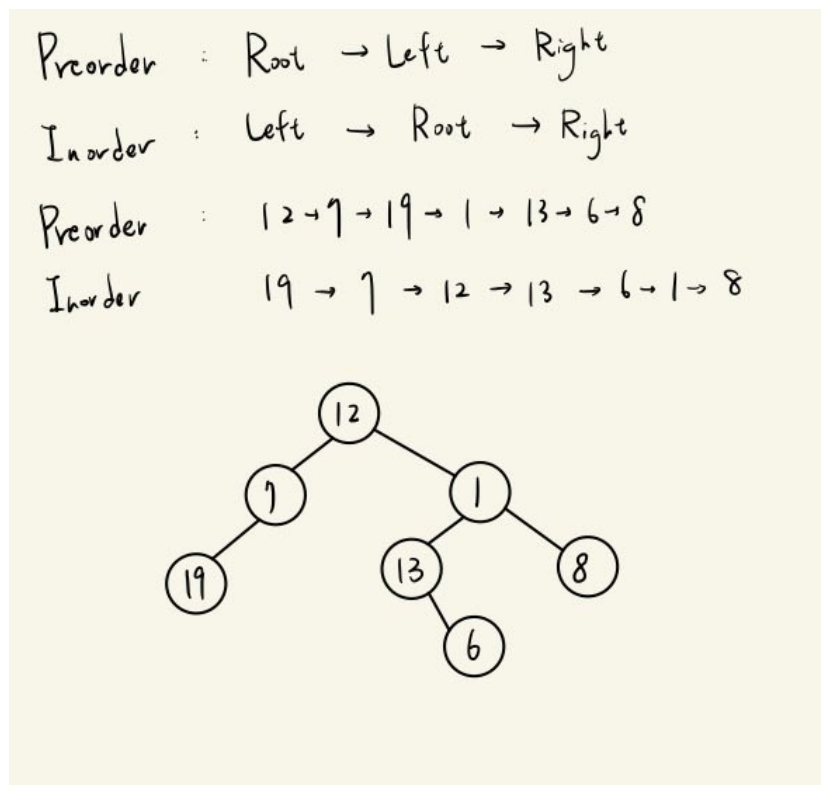


Figure 3: Binarytree

(c)

2 Recursion Problem

(a)

(b)

3 AVL Tree

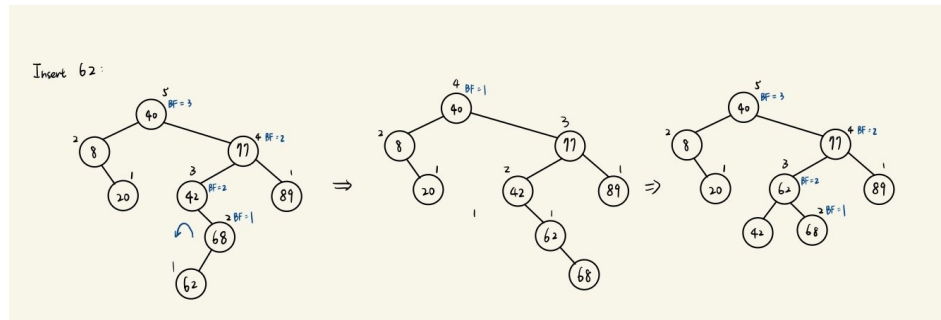


Figure 4: Binarytree

(a)

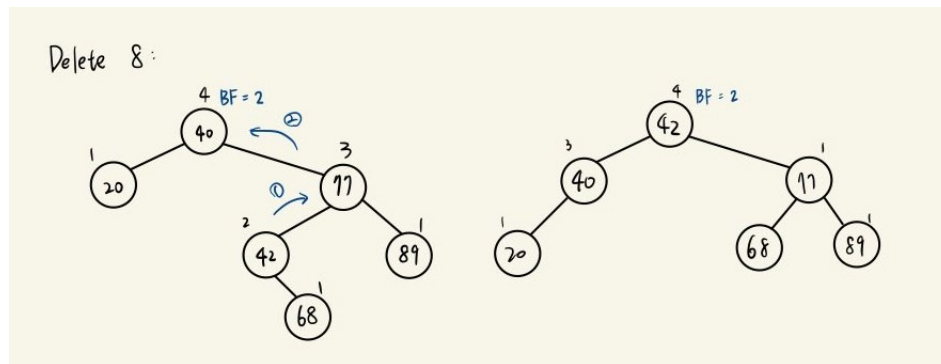


Figure 5: Binarytree

(b)

(c) The minimum height is $\lceil (\log_2(n+1)) \rceil$.

Since AVL Tree must follow the rule $|left_height - right_height| \leq 1$, the worst case of AVL Tree is Fibonacci Tree. Fibonacci tree gives us $T(0) = 0, T(1) = 1, T(h) = T(h-1) + T(h-2) + 1$, where h means the tree's height, and $T(h)$ means the number of nodes in the h -height Fibonacci Tree. In this recurrence relation, we can easily compute the general function $T(h) = \frac{(\phi^{h+2} - \varphi^{h+2})}{\sqrt{5}} - 1$, where ϕ is golden ratio and φ is its conjugate.

In this case, we want to find h such that $T(h) = 493$. We can set $\varphi^{h+2} = \epsilon$ since it's much more smaller than ϕ^{h+2} . Then the equation will become

$$T(h) = \frac{\phi^{h+2}}{\sqrt{5}} - \epsilon - 1 = 493$$

After solving the equation, we have $h = \lfloor \log_{\phi} \sqrt{5}(494 + \epsilon) - 2 \rfloor$. After computing the $\lfloor \log_{\phi} \sqrt{5}(494 + \epsilon) - 2 \rfloor = 12$ and $\lceil (\log_2(494)) \rceil = 9$, the possible heights is 9, 10, 11, 12.

4 Parentheses Balance Problem

- (a)
- (b)
- (c)

5 Complexity Analysis

- (a) Consider the recurrence relation

$$\begin{cases} a_n &= a_{n-2} + n \\ a_1 &= k_1(\text{constant}) \\ a_0 &= k_0(\text{constant}) \end{cases}$$

If n is even, we have $a_n = a_0 + \sum_{k=1}^{\frac{n}{2}} k = a_0 + \frac{1}{2} \cdot \frac{n}{2} \cdot \frac{n+2}{2}$. Hence the complexity is $O(n^2)$.

- (b) Consider the recurrence relation

$$\begin{cases} a_n &= 2a_{n-1} + n \\ a_1 &= k(\text{constant}) \end{cases}$$

We have the general $a_n = 2^{n-1}(a_1 + 3)$, which means the complexity is $O(2^n)$

- (c) Consider the recurrence relation

$$\begin{cases} (a_n + \frac{1}{4}) &= 2(a_{n-1} + \frac{1}{4}) + 3(a_{n-2} + \frac{1}{4}) \\ a_1 &= 1 \\ a_2 &= 1 \end{cases}$$

After solving its characteristic equation $r^2 = 2r + 3$, we get two roots 3, -1. The general $a_n = C \cdot (3)^n + D \cdot (-1)^n$. Hence the complexity is $O(3^n)$

- (d) Consider the recurrence relation

$$\begin{cases} a_n &= 8a_{\frac{n}{4}} + \log n \\ a_1 &= k(\text{constant}) \end{cases}$$

The recurrence relation will give us $a_n = 8^{\log_4 n} a_1 + O(\log n) = n^{\frac{3}{2}} + O(\log n)$. Since we know polynomial is growing faster than log, $T(n)$ has time complexity $O(n^{\frac{3}{2}})$.

- (e) $T(n) = T(\sqrt{n}) + 1$, assume that when m -th recurrence, $n^{\frac{1}{2^m}} \approx 1 + 10^{-6}$. In the m -th recurrence we have

$$T(n) = T(n^{\frac{1}{2^m}}) + m \approx m + 1$$

Compute m :

$$\log n = 2^m \log(1 + 10^{-6}) \Rightarrow m = \log_2\left(\frac{\log n}{\log(1 + 10^{-6})}\right)$$

Hence, the complexity is $O(\log \log n)$

- (f) $T(n) = 3T(\sqrt{n}) + 1$, assume that when m -th recurrence, $n^{\frac{1}{2^m}} \approx 1 + 10^{-6}$. In the m -th recurrence we have

$$T(n) = 3^m \cdot T(n^{\frac{1}{2^m}}) + m \approx m + 3^m$$

Compute m :

$$\log n = 2^m \log(1 + 10^{-6}) \Rightarrow m = \log_2\left(\frac{\log n}{\log(1 + 10^{-6})}\right)$$

Then 3^m :

$$3^{\log_2\left(\frac{\log n}{\log(1 + 10^{-6})}\right)} = \log_2 3 \frac{\log n}{\log(1 + 10^{-6})}$$

Hence, the complexity is $O(\log n)$

6 Problem 6